Sec 4.1

Systems of equations

Just like we say an ordered pair (x,y) is a solution to $y=ax+c$ when (x,y) keep or satisfy the equation we can require that (x,y) satisfy more than 1 equation

Def

Suppose that (x,y) satisfy both of the following equations

$$\begin{cases} ax+by=c \\ dx+ey=f \end{cases}$$

Then we say the ordered pair (x,y) solve the system of equations above.

Ex

Check wether $(3,3)$ and $(4,2)$ are solutions to the system

$$\begin{cases} x+y=6 \\ 2x-5y=-2 \end{cases}$$

Substitute ordered pairs in.

for $(3,3)$ $x=3$ $y=3$

so

$3+3=6$ ✓

$2(3)-5(3)=6-15=-9 \neq -2$ ✗

for $(4,2)$

$4+2=6$ ✓

$2(4)-5(2)=8-10=-2$ ✓

So $(4,2)$ is a solution.

RQ Question 1 in class

Lets think about

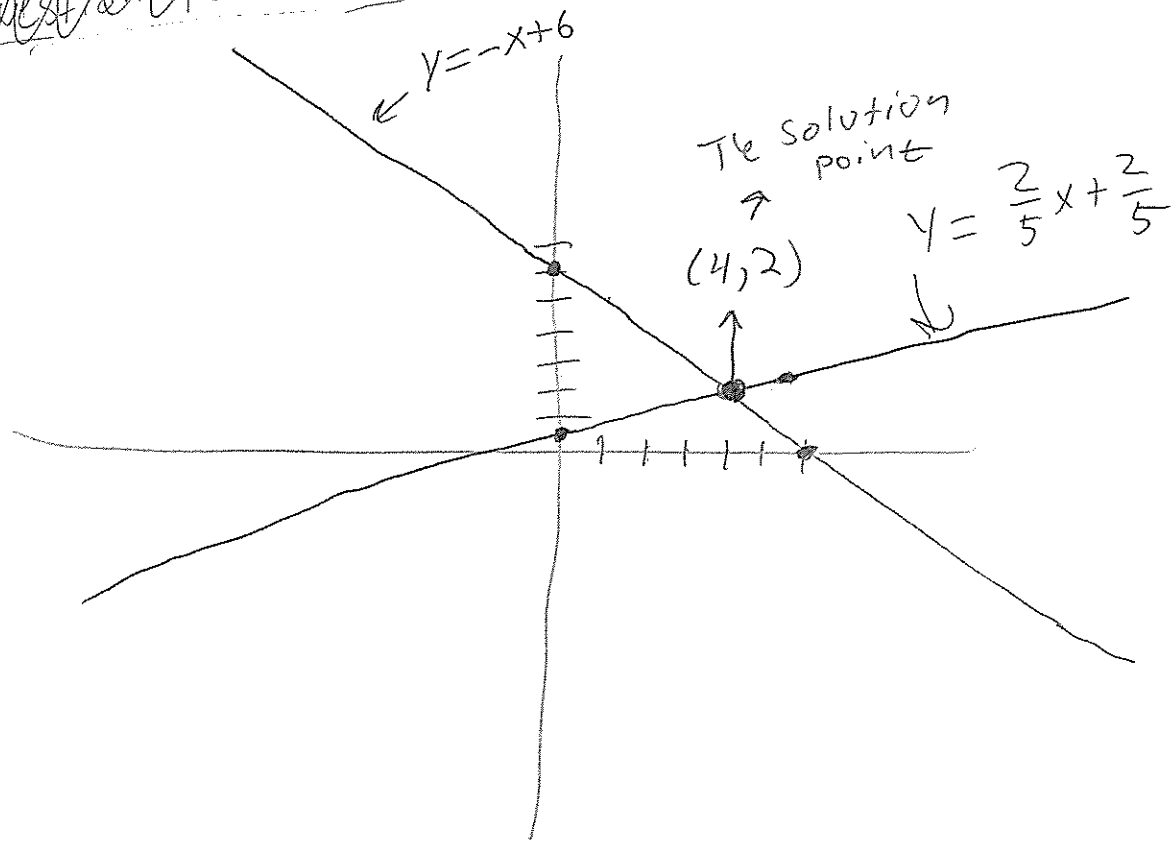
$$\begin{cases} x+y=6 \\ 2x-5y=-2 \end{cases}$$

These are two lines →

$$\begin{cases} y=-x+6 \\ y=\frac{2}{5}x+\frac{2}{5} \end{cases}$$

So if a point (x,y) satisfies each equation it must lie on <u>both</u> lines.

Lets graph them.

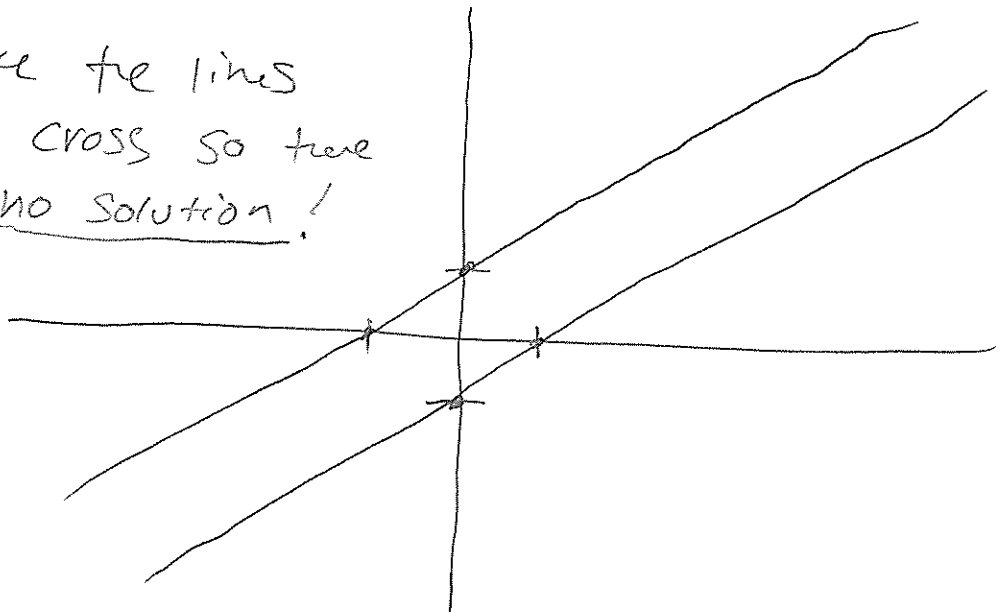


Note here there is only one solution, te Point where te lines intercect.

What if we have parallel lines like

$$\begin{cases} y = x + 1 \\ y = x - 1 \end{cases}$$

Notice te lines never cross so there is no solution!



What about the lines

$$\begin{cases} y = 3x + 2 \\ 2y = 6x + 4 \end{cases}$$

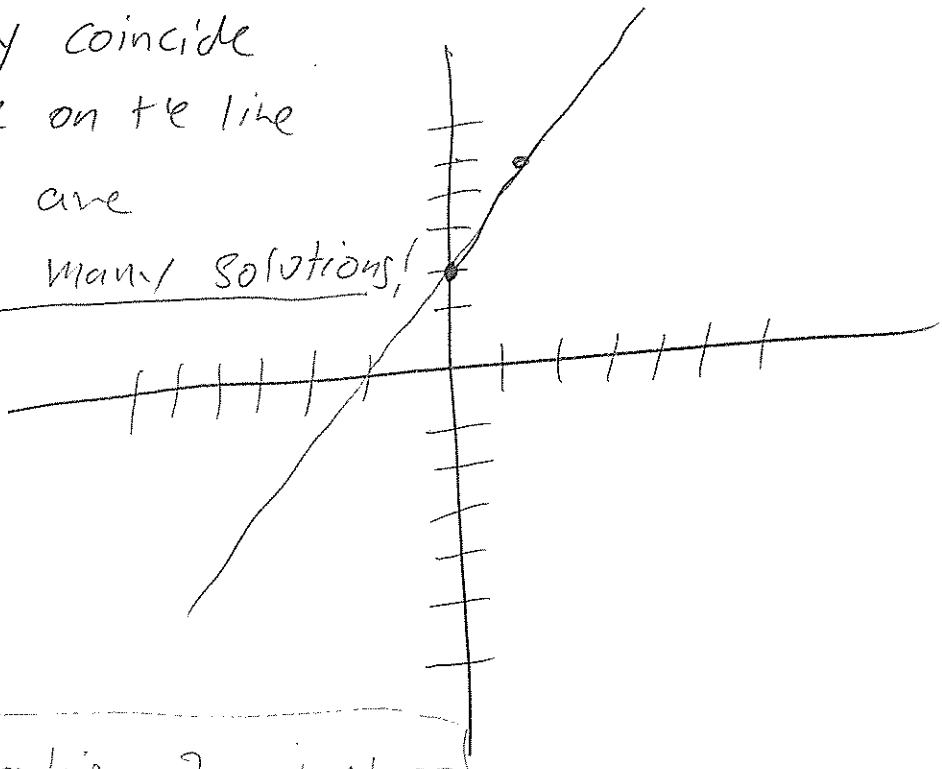
Notice $2y = 6x + 4$ is the same as $y = 3x + 2$ if we divide both sides by 2

$$\frac{2y}{2} = \frac{6x+4}{2}$$

$$\Rightarrow y = 3x + 2$$

So they are the same line!

Thus they coincide everywhere on the line and there are Infinitely many solutions!



RQ Question 2 in class

Solving systems of equations

Method one

Substitution

1) Solve one of the equations for one variable in terms of the other.

2) Plug in the expression from step 1 into the other equation

3) Solve for the equation in step 2.

4) Substitued the solution from step 3 into the first equation and then solve for the last variable

Ex

$$\begin{cases} x + 2y = -4 \\ 5x - y = 13 \end{cases}$$

① I'll choose to solve the first equation for y interms of x

$$x + 2y = -4$$
$$-x \qquad -x$$
$$\frac{2y}{2} = \frac{-4 - x}{2}$$
$$\boxed{y = -2 - \tfrac{1}{2}x}$$

② since $y = -2 - \frac{1}{2}x$ I can replace y in the second equation with $-2 - \frac{1}{2}x$

$5x - y = 13$

$\downarrow$

$= -2 - \frac{1}{2}x$

③ so

$5x - (-2 - \frac{1}{2}x) = 13$

distribute

$5x + 2 + \frac{1}{2}x = 13$

$-2 \qquad -2$

← Now everything is in terms of x so I can solve for x

$5x + \frac{1}{2}x = 11$

$\frac{10}{2}x + \frac{1}{2}x = 11$

$2(\frac{11}{2}x) = (11)2$

$\frac{11x}{11} = \frac{22}{11}$

$\boxed{x = 2}$ ← Now I put 2 in for x in the first equation

④ $x + 2y = -4$ and $x = 2$ so:

$2 + 2y = -4$

$-2 \quad -2$

$2y = -6$

$\frac{2y}{2} = \frac{-6}{2}$

$\boxed{y = -3}$